System Design and Data Dictionary

Entity relationships architecture and workflow diagrams

Group 1 MIT 820 Application 1.0.4 Document 1.1 23 October 2026

This design document links the CivicFlow 1.0.4 application to its architecture, record relationships, workflow and permission decisions. The diagrams provide separate views to keep the 21-table schema readable. All columns and declared foreign keys appear in the dictionary. Dashed diagram links identify logical associations that are not database foreign keys.

1 Notation and data boundaries

PK denotes a primary key, FK a declared foreign key and UNIQUE a uniqueness constraint. A 1:M label means one to many. A solid relationship line represents a declared association shown in that view. Some repeated actor/uploader references appear as FK labels without an additional crossing line; the complete foreign-key table is authoritative. Optional fields are nullable. A logical reference is enforced by application behaviour without a declared database FK.

The citizen owns a request, the selected service supplies its default department, and current routing may change the processing department. The optional assignee is a staff user. Request versions guard concurrent edits. Templates and outbox messages retain distinct identities so a later template change does not rewrite queued content.

2 Architecture and diagrams

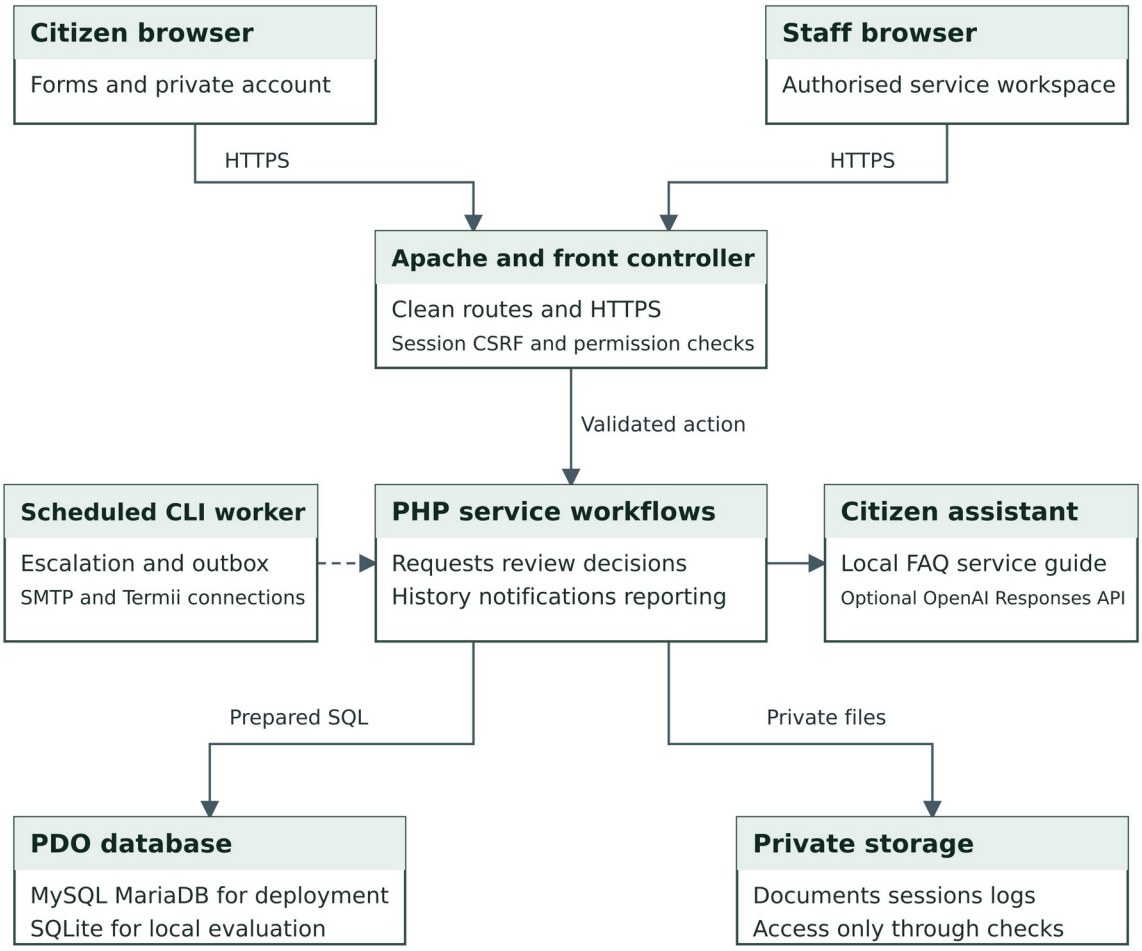


Figure D1 Application architecture

Source: Project implementation and system design for release 1.0.4.

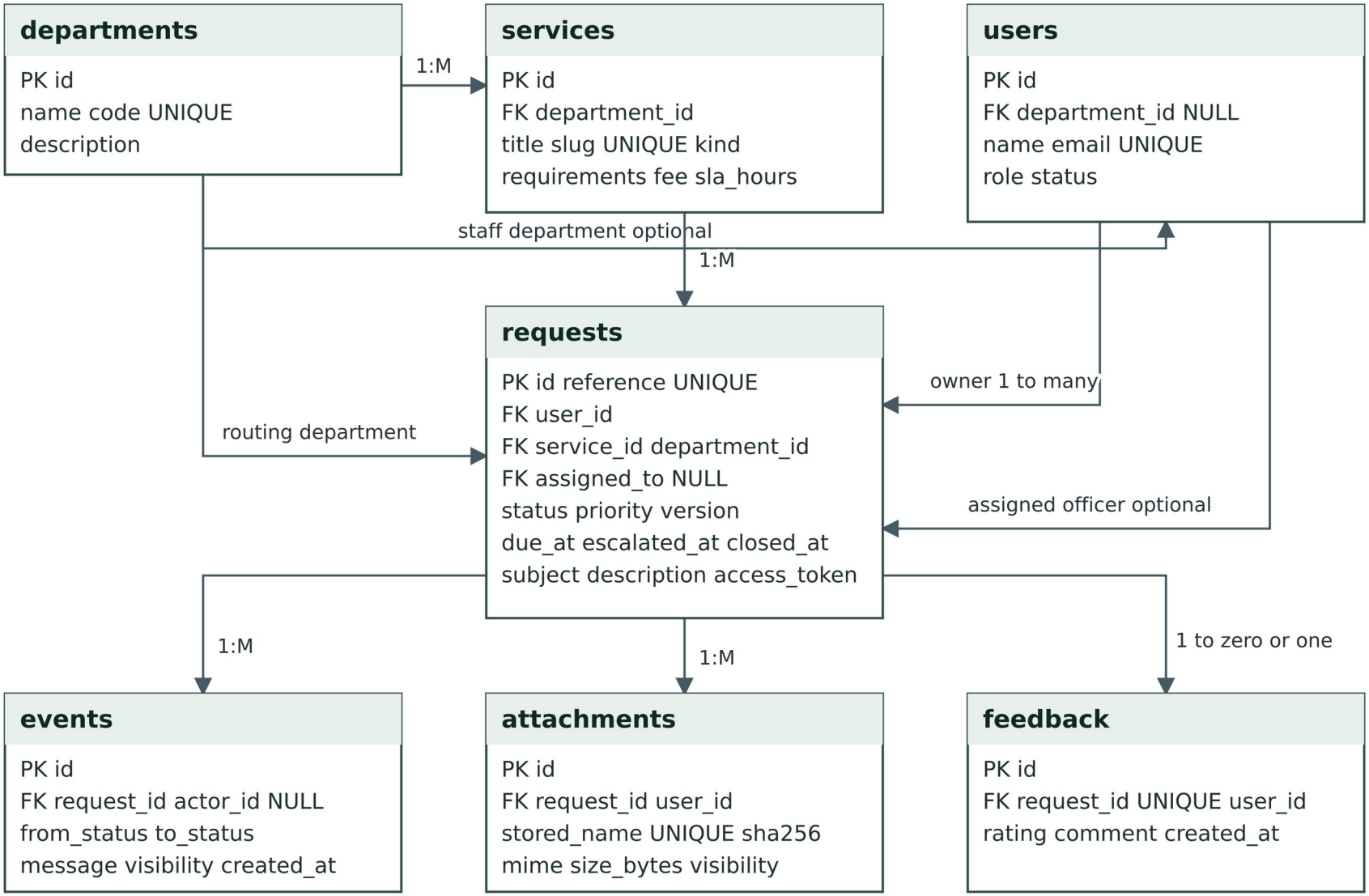


Figure D2 Core service entity relationships

Source: Project implementation and system design for release 1.0.4.

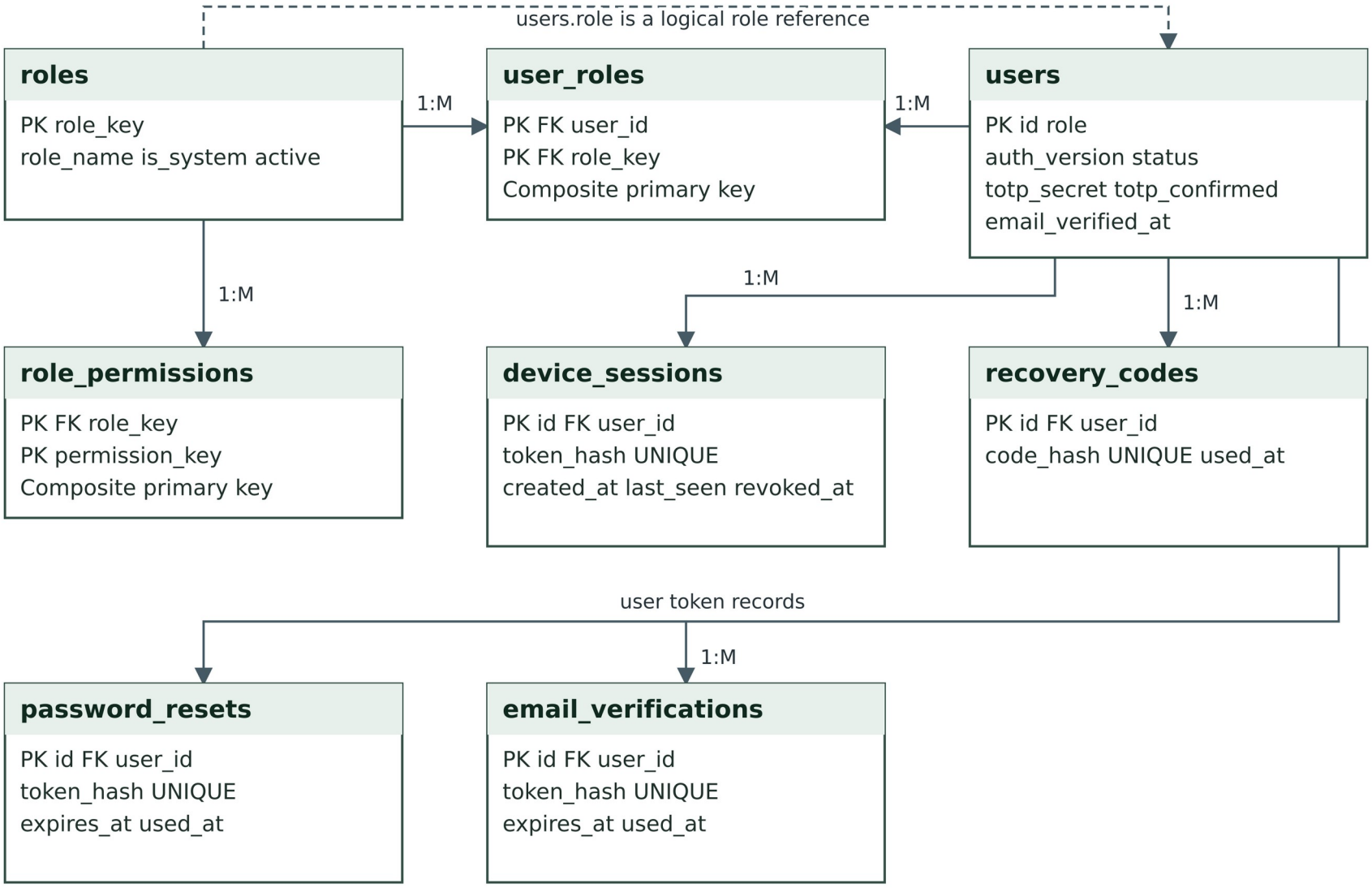


Figure D3 Identity and permission relationships

Source: Project implementation and system design for release 1.0.4.

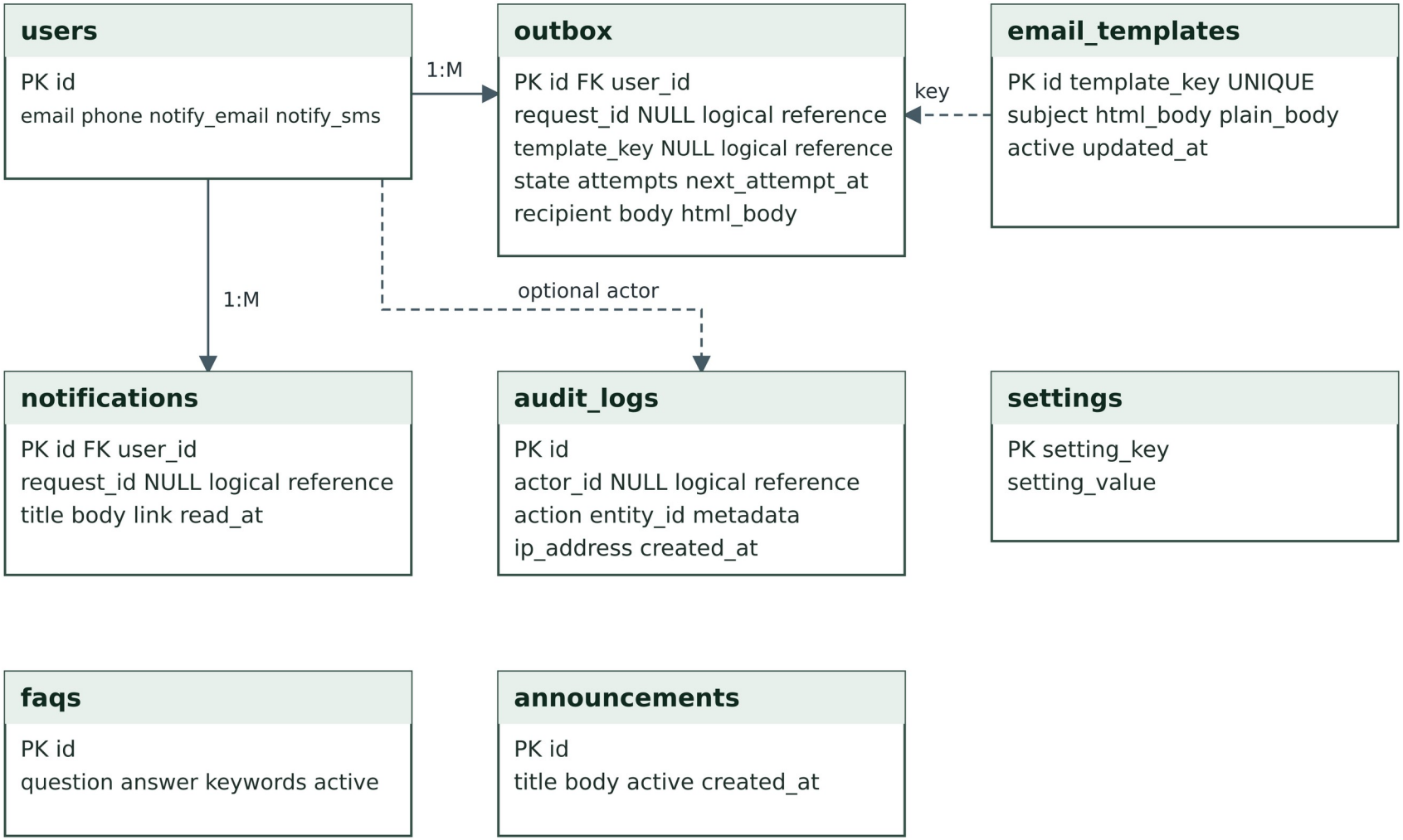


Figure D4 Communication and workspace relationships

Source: Project implementation and system design for release 1.0.4.

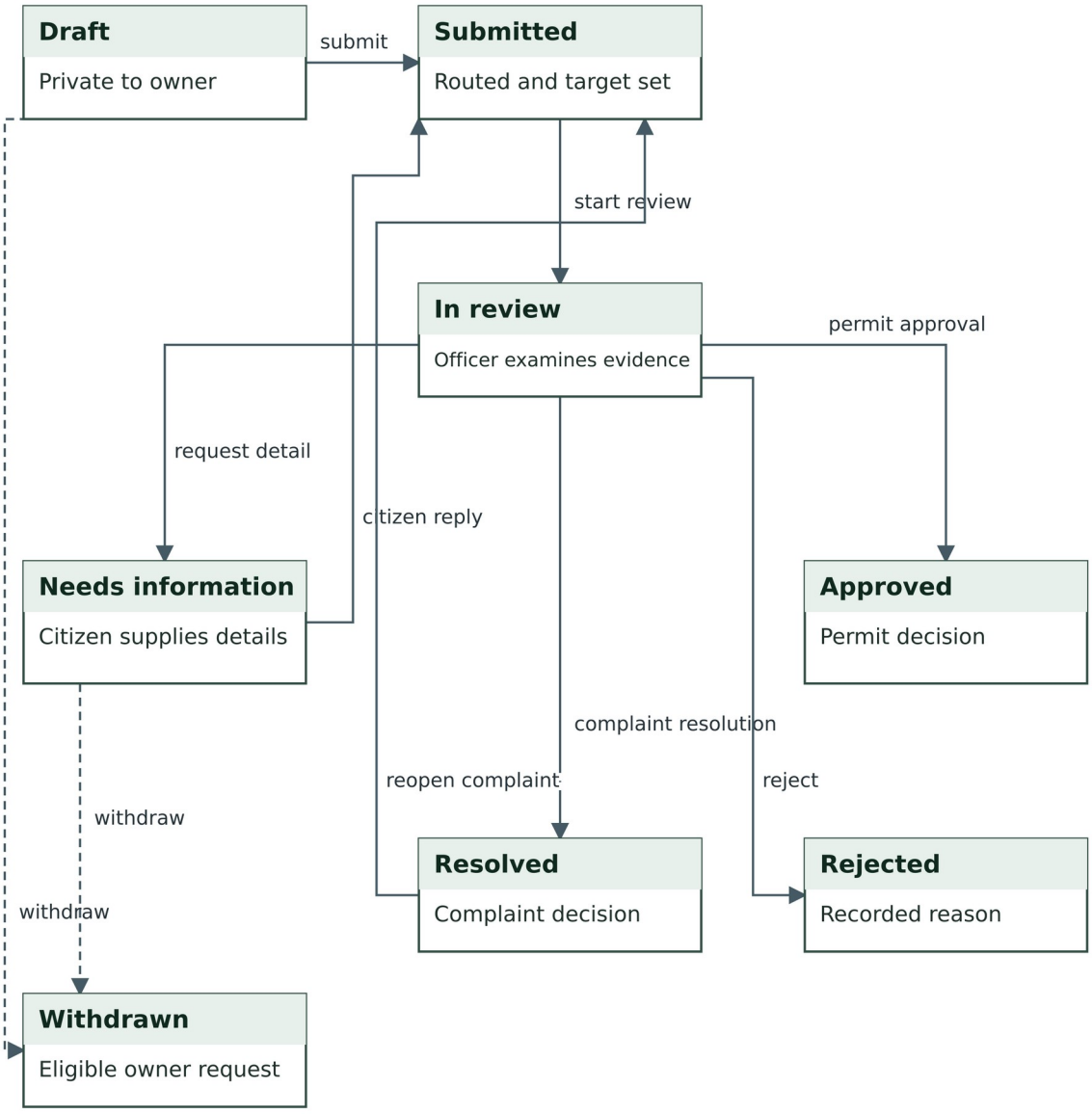


Figure D5 Complaint and permit lifecycle

Source: Project implementation and system design for release 1.0.4.

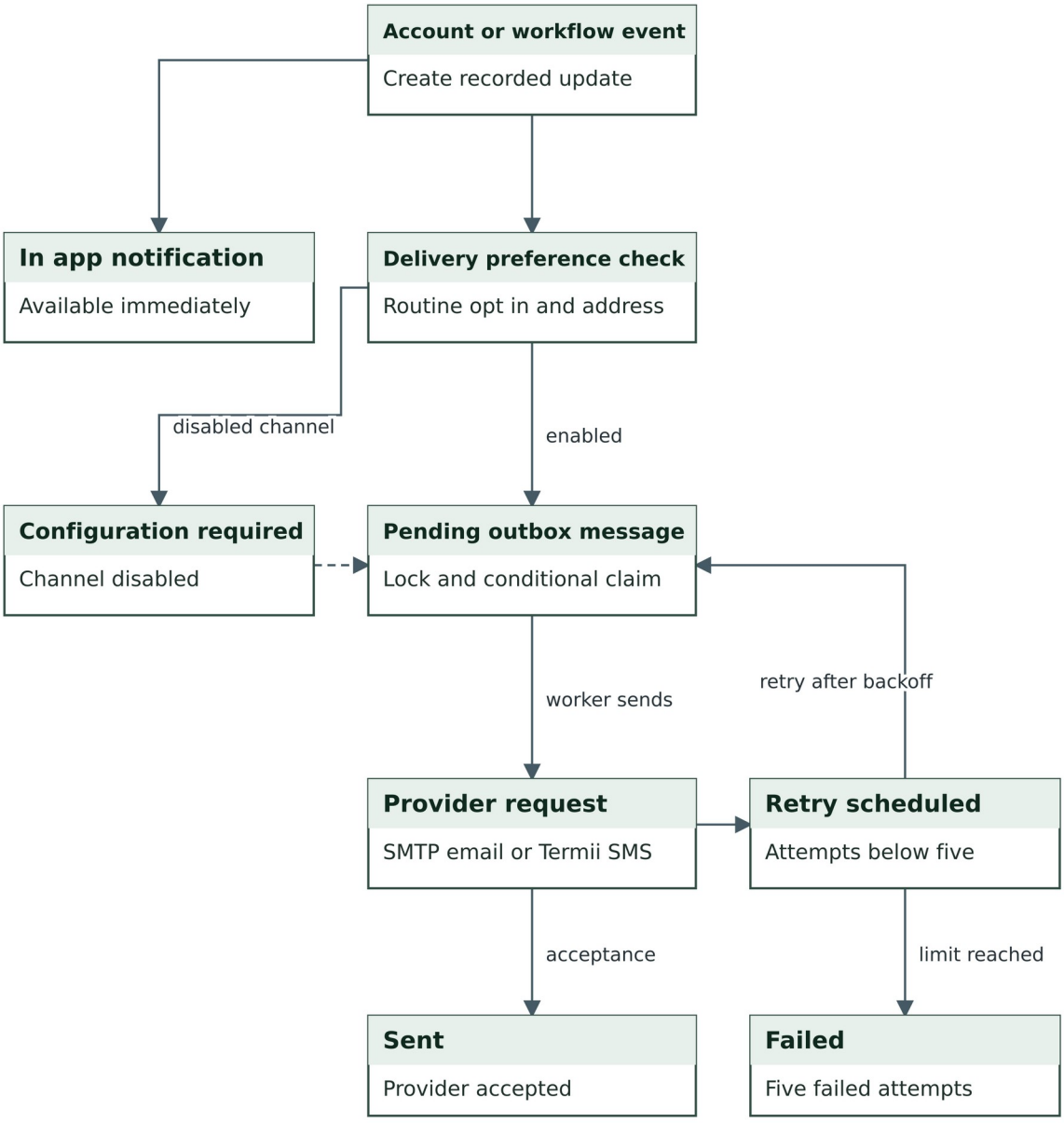


Figure D6 Delivery and retry workflow

Source: Project implementation and system design for release 1.0.4.

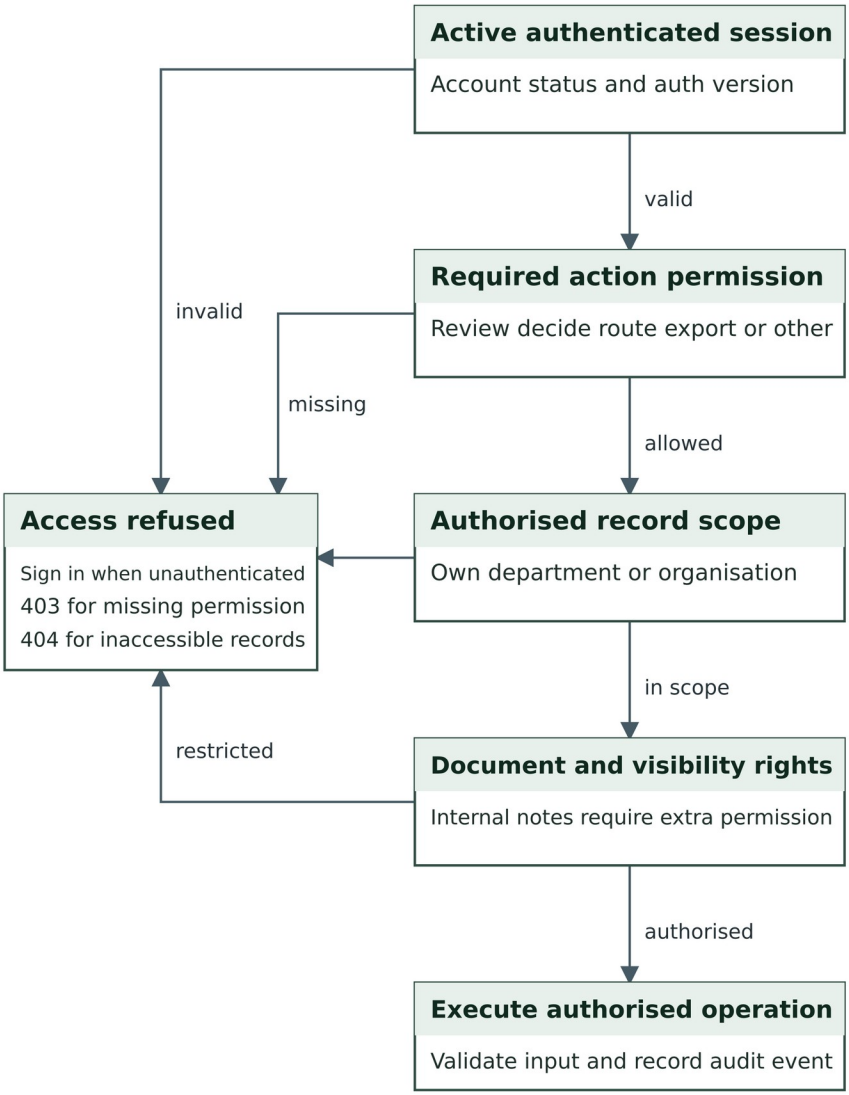


Figure D7 Server side access decision

Source: Project implementation and system design for release 1.0.4.

3 Complete schema data dictionary

The following tables list the declared columns and database foreign keys from the schema used by the source release. INTEGER primary identifiers use automatic numbering appropriate to the selected engine. The SQL schema remains the source for complete uniqueness and default-expression syntax. Operational permissions are defined in application code and role-permission records.

departments

Column	Type	Declared constraints
id	INTEGER	PK; Required
name	VARCHAR(140)	Required
code	VARCHAR(20)	Required
description	TEXT	Required

services

Column	Type	Declared constraints
id	INTEGER	PK; Required
department_id	INTEGER	Required; FK departments.id
title	VARCHAR(180)	Required
slug	VARCHAR(100)	Required
kind	VARCHAR(20)	Required
description	TEXT	Required
requirements	TEXT	Required
fee	DECIMAL(12,2)	Required; Default 0
sla_hours	INTEGER	Required; Default 72
active	INTEGER	Required; Default 1

roles

Column	Type	Declared constraints
role_key	VARCHAR(50)	PK; Required
role_name	VARCHAR(100)	Required

Column	Type	Declared constraints
description	TEXT	Required
is_system	INTEGER	Required; Default 0
active	INTEGER	Required; Default 1

role_permissions

Column	Type	Declared constraints
role_key	VARCHAR(50)	PK composite; Required; FK roles.role_key
permission_key	VARCHAR(100)	PK composite; Required

users

Column	Type	Declared constraints
id	INTEGER	PK; Required
name	VARCHAR(140)	Required
email	VARCHAR(190)	Required
phone	VARCHAR(25)	Required; Default ''
password_hash	VARCHAR(255)	Required
role	VARCHAR(50)	Required; Default 'citizen'
department_id	INTEGER	Nullable; FK departments.id
status	VARCHAR(20)	Required; Default 'active'
email_verified_at	DATETIME	Nullable
notify_email	INTEGER	Required; Default 1
notify_sms	INTEGER	Required; Default 0
auth_version	INTEGER	Required; Default 1
totp_secret	TEXT	Nullable
totp_confirmed	INTEGER	Required; Default 0
created_at	DATETIME	Required

user_roles

Column	Type	Declared constraints
user_id	INTEGER	PK composite; Required; FK users.id
role_key	VARCHAR(50)	PK composite; Required; FK roles.role_key

email_templates

Column	Type	Declared constraints
id	INTEGER	PK; Required
template_key	VARCHAR(60)	Required
name	VARCHAR(140)	Required
subject	VARCHAR(180)	Required
html_body	TEXT	Required
plain_body	TEXT	Required
active	INTEGER	Required; Default 1
updated_at	DATETIME	Required

device_sessions

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
token_hash	VARCHAR(64)	Required
ip_address	VARCHAR(64)	Required
user_agent	VARCHAR(250)	Required
created_at	DATETIME	Required
last_seen	DATETIME	Required
revoked_at	DATETIME	Nullable

recovery_codes

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
code_hash	VARCHAR(64)	Required
used_at	DATETIME	Nullable

requests

Column	Type	Declared constraints
id	INTEGER	PK; Required
reference	VARCHAR(50)	Required
access_token	VARCHAR(64)	Required
user_id	INTEGER	Required; FK users.id
service_id	INTEGER	Required; FK services.id
department_id	INTEGER	Required; FK departments.id
assigned_to	INTEGER	Nullable; FK users.id
kind	VARCHAR(20)	Required
subject	VARCHAR(180)	Required
description	TEXT	Required
location	VARCHAR(200)	Required; Default ''
state_name	VARCHAR(60)	Required; Default ''
organisation	VARCHAR(180)	Required; Default ''
priority	VARCHAR(20)	Required; Default 'normal'
status	VARCHAR(30)	Required
due_at	DATETIME	Nullable
escalated_at	DATETIME	Nullable
closed_at	DATETIME	Nullable
version	INTEGER	Required; Default 1

Column	Type	Declared constraints
created_at	DATETIME	Required
updated_at	DATETIME	Required

events

Column	Type	Declared constraints
id	INTEGER	PK; Required
request_id	INTEGER	Required; FK requests.id
actor_id	INTEGER	Nullable; FK users.id
from_status	VARCHAR(30)	Required; Default ''
to_status	VARCHAR(30)	Required; Default ''
message	TEXT	Required
visibility	VARCHAR(20)	Required; Default 'public'
created_at	DATETIME	Required

attachments

Column	Type	Declared constraints
id	INTEGER	PK; Required
request_id	INTEGER	Required; FK requests.id
user_id	INTEGER	Required; FK users.id
original_name	VARCHAR(200)	Required
stored_name	VARCHAR(100)	Required
mime	VARCHAR(100)	Required
size_bytes	INTEGER	Required
sha256	VARCHAR(64)	Required
visibility	VARCHAR(20)	Required; Default 'public'
created_at	DATETIME	Required

notifications

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
request_id	INTEGER	Nullable
title	VARCHAR(180)	Required
body	TEXT	Required
link	VARCHAR(250)	Required
read_at	DATETIME	Nullable
created_at	DATETIME	Required

outbox

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
request_id	INTEGER	Nullable
channel	VARCHAR(15)	Required
recipient	VARCHAR(190)	Required
subject	VARCHAR(180)	Required
body	TEXT	Required
html_body	TEXT	Nullable
template_key	VARCHAR(60)	Nullable
state	VARCHAR(30)	Required; Default 'pending'
attempts	INTEGER	Required; Default 0
next_attempt_at	DATETIME	Nullable
sent_at	DATETIME	Nullable
last_error	VARCHAR(250)	Required; Default ''
provider_id	VARCHAR(150)	Required; Default ''

Column	Type	Declared constraints
created_at	DATETIME	Required

settings

Column	Type	Declared constraints
setting_key	VARCHAR(100)	PK; Required
setting_value	TEXT	Required

password_resets

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
token_hash	VARCHAR(64)	Required
expires_at	DATETIME	Required
used_at	DATETIME	Nullable

email_verifications

Column	Type	Declared constraints
id	INTEGER	PK; Required
user_id	INTEGER	Required; FK users.id
token_hash	VARCHAR(64)	Required
expires_at	DATETIME	Required
used_at	DATETIME	Nullable

audit_logs

Column	Type	Declared constraints
id	INTEGER	PK; Required
actor_id	INTEGER	Nullable

Column	Type	Declared constraints
action	VARCHAR(100)	Required
entity_id	INTEGER	Nullable
metadata	TEXT	Required
ip_address	VARCHAR(64)	Required
created_at	DATETIME	Required

feedback

Column	Type	Declared constraints
id	INTEGER	PK; Required
request_id	INTEGER	Required; FK requests.id
user_id	INTEGER	Required; FK users.id
rating	INTEGER	Required
comment	TEXT	Required
created_at	DATETIME	Required

faqs

Column	Type	Declared constraints
id	INTEGER	PK; Required
question	VARCHAR(220)	Required
answer	TEXT	Required
keywords	VARCHAR(250)	Required
active	INTEGER	Required; Default 1

announcements

Column	Type	Declared constraints
id	INTEGER	PK; Required
title	VARCHAR(180)	Required

Column	Type	Declared constraints
body	TEXT	Required
active	INTEGER	Required; Default 1
created_at	DATETIME	Required

4 Declared foreign key register

Referencing column	Referenced primary key
services.department_id	departments.id
role_permissions.role_key	roles.role_key
users.department_id	departments.id
user_roles.role_key	roles.role_key
user_roles.user_id	users.id
device_sessions.user_id	users.id
recovery_codes.user_id	users.id
requests.assigned_to	users.id
requests.department_id	departments.id
requests.service_id	services.id
requests.user_id	users.id
events.actor_id	users.id
events.request_id	requests.id
attachments.user_id	users.id
attachments.request_id	requests.id
notifications.user_id	users.id
outbox.user_id	users.id
password_resets.user_id	users.id
email_verifications.user_id	users.id
feedback.user_id	users.id
feedback.request_id	requests.id

5 Logical reference register

Field	Application relationship
users.role	Primary role key in roles, without a database FK.
outbox.request_id	Optional service request link, without a database FK.
outbox.template_key	Optional template key retaining the queued message provenance.
notifications.request_id	Optional service request link, without a database FK.
audit_logs.actor_id	Optional user actor link, without a database FK.

6 Diagram source files

Each diagram is included as a high-resolution PNG and editable SVG. The source architecture/workflow document also provides Mermaid definitions. The field dictionary and SQL references resolve any detail omitted from the selected diagram lines. Diagram files are design models, not screenshots or proof of a hosted deployment.

7 Additional formal and deployment views

The Architecture Design Document provides the architectural alternatives, decisions, threat considerations and single-instance deployment assumptions. The diagram collection adds a UML-style use case view and submission sequence, a deployment topology and two interface wireframes. PNG files provide ready-to-display images; SVG and Mermaid sources remain editable. Wireframes are design views, not evidence of a live browser test.

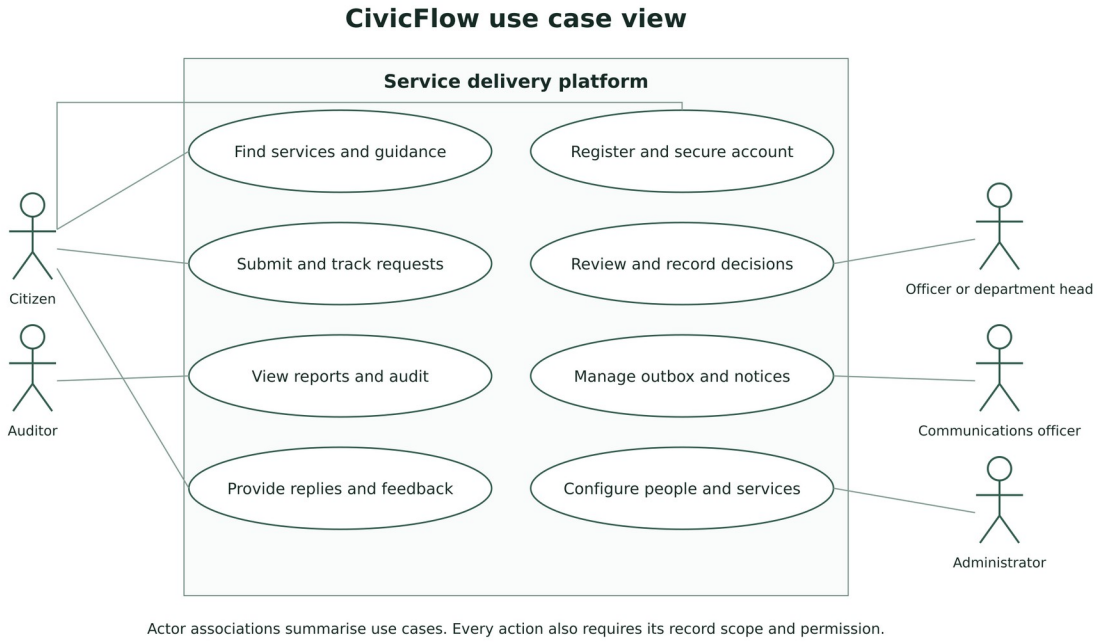


Figure 7.1 Principal actors and use cases

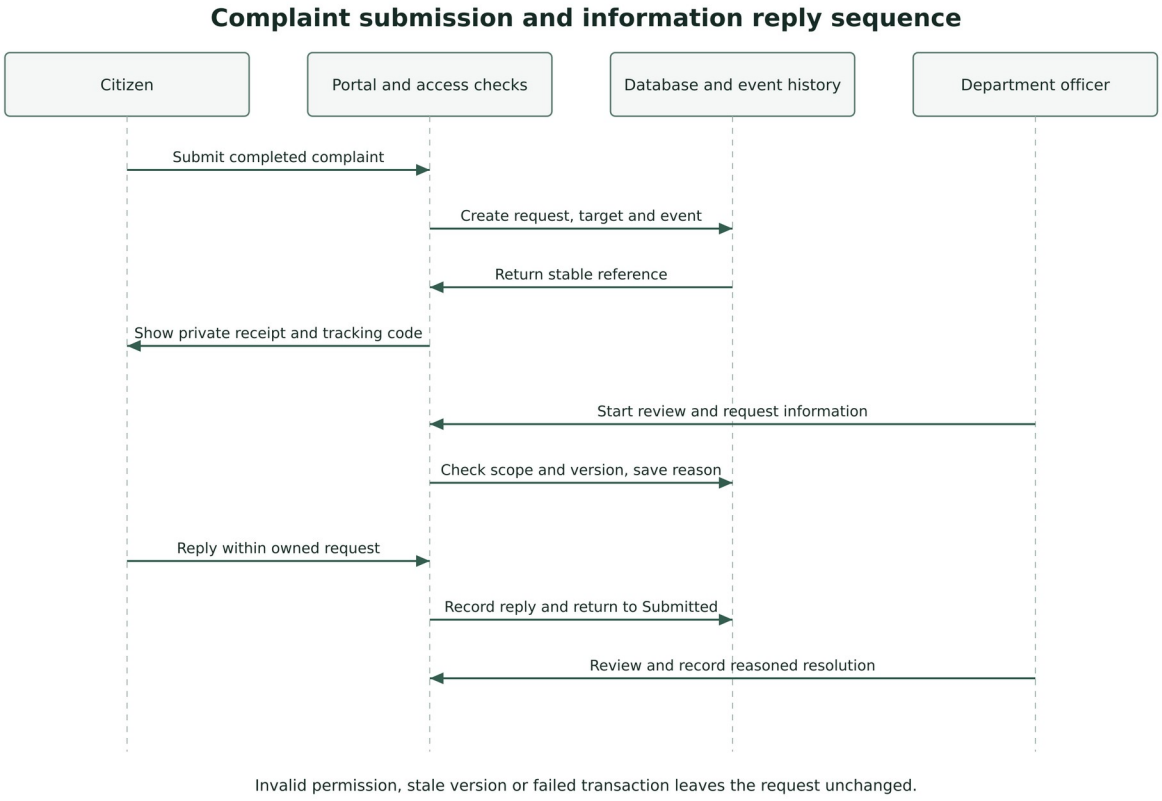


Figure 7.2 Submission and information reply sequence

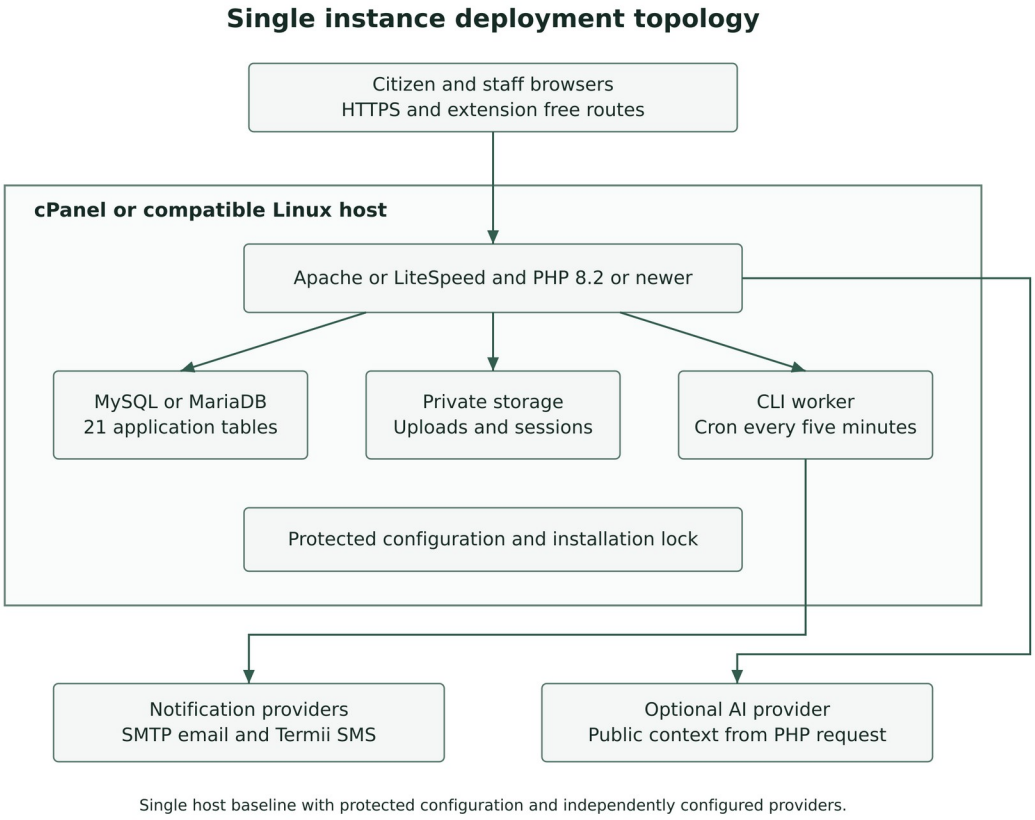


Figure 7.3 Single instance deployment topology